MOCK EXAMS
MATHEMATICS 2 & 1
Essay and Objectives
2 hours

2&1

Name:.....

School: .....

Index Number:.....

Date: .....

Signature: .....

Mock Examination

MATHEMATICS 2 & 1

2 hours

Do **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions carefully. Write your* **name** *and* **index number** *in the spaces provided above.*

The examination consist of **two** papers: Paper **1** will be Objective Test and Paper **2** will be Essay.

1. **PAPER 1 (OBJECTIVE)**
   This consist of **40** compulsory objective questions and will last for **1 hour**. This will carry **40 marks**.

2. **PAPER 2 (ESSAY)**
   This consist of **six** questions and candidate is expected to answer four out of **six** questions. The Paper will last **1 hour** and will carry **60 marks**.

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PAPER 2 1 hour

ESSAY

Answer **four** *questions only.*

All question carry equal marks

All working must be clearly shown. Marks will **not** *be awarded for correct answers with out corresponding working.*

1. (a) The ratio of the ages of Francis Odoom and Hilda Aboagye is 2: 7 respectively. Hilda Aboagye is 15 years older than Francis Odoom. Let x represent the age of Hilda Aboagye

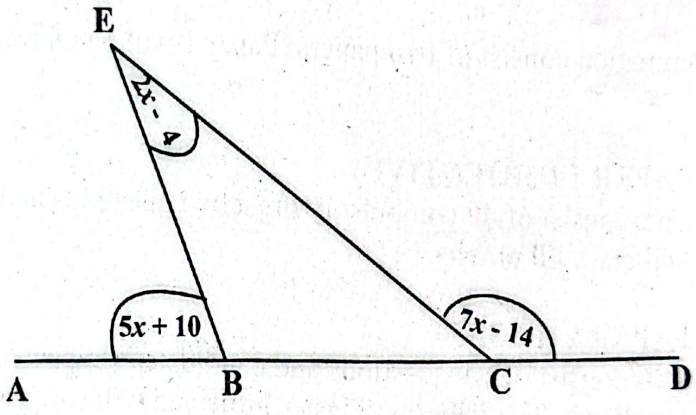
   (i) Write down an algebraic expression representing the age of Francis Odoom

   (ii) Find the value of x.

   (iii) Find the sum of the ages of Hilda Aboagye and Francis Odoom.

   (b) Calculate the simple interest on GH¢ 5,000 at 2% simple interest rate per month for 2 years.

   (c) In the diagram, ABCD is a straight line, $\angle ABE = (15x + 10)^\circ$ $\angle DCE = (7x - 14)^\circ$ and $\angle BEC = (2x - 4)^\circ$. Find the value of x.



2 (a) Factorize completely $(2x + y)(a - b) - x(a - b)$

   (b) Ahava - Dorothy paid GH¢ 280.00 for a blender after a discount of 20% was allowed.

   (i) Calculate the discount

   (ii) If she decide to sell the item at the marked price, at 12 monthly instalment, how much will one pay at the end of each month for the blender?

   (c) The marks of 5 students out of 25 was supposed to be used to draw a pie chart.

   Find;

   (i) the total marks of the 5 students

   (ii) the value of a, b, c, d, e from the table below

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Names	Marks	Angel of sector
Thelma	24	180
Vera	a	60
Chelsea	d	b
Daniel	4	e
Mabel	c	75

3. (a) A model designed by Henry and Ato Company Limited to determine the amount to pay for purchasing a number of rice cooker gave as the table below.

Number of rice cooker (x)	1	2	3	4	5
Price in GH¢ (y)	200		300		400

(i) Deduce a formulae for finding the cost of rice cooker (y) for a number of rice cooker (x)

(ii) Copy and complete the table

(iii) Using a scale of 4 cm to 1 unit on the x - axis and 2 cm to 50 units on the y - axis, draw the graph of number of rice cooker (x) against the price (y)

(b) It took Seshie 55 minutes to walk from home to school. If he got to school at 7 : 12 am , when did he leave home to school?

4. (a) The point M is on the same level ground as the foot of a vertical pole. The angle of elevation of the top of the pole from M is 55°. If M is 10m from the foot of the pole, calculate the height of the pole using a diagram. [Take tan 55° = 1.428]

(b) Given that $\mathbf{u} = \begin{pmatrix} 6 \\ 5 \end{pmatrix}$ $\mathbf{v} = \begin{pmatrix} -6 \\ 4 \end{pmatrix}$ find $\frac{1}{3}$ $(\mathbf{u} - \frac{1}{2}\mathbf{v})$

(c) Find the truth set of $\frac{8x - 3}{2} - \frac{10x - 6}{4} - \frac{12x - 1}{12} \geq 4\frac{1}{12}$

5. (a) Using a ruler and a pair of compasses only, construct

(i) triangle ABC such that |AB| = 12 cm, |AC| = 8 cm and angle BAC = 30°.

(ii) a perpendicular from c to meet AB at M.

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(b) Measure;
(i) angle ABC
(ii) |CM|

(c) Calculate the area of triangle ABC

6. (a) The table below shows the probabilities of five drinks liked by 30 students

Drinks,	Fanta	Coke	Malt	Sprite	Pepsi
Probability	0.2	0.1	0.4	0.15	

(i) Find the probability of students who preferred Pepsi
(ii) Find the umber of students who liked malt.

(b) Evaluate leaving your answer in standard form: $(4 \times 10^{3}) \times (6 \times 10^{-5})$

(c) Simplify $6\frac{1}{3} - 3\frac{2}{5} + 2\frac{3}{8}$

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Answer all the questions

Each question is followed by **four** *options lettered A to D. Find the correct option for* **each** *question and shade* **in pencil** *on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An examples is given below:*

If $3n + 2 = 8$, find the value of n.
A. 10
B. 6
C. 3
D. 2

The correct answer is 2, which is lettered D and therefore answer spaces D would be shaded. [A] [B] [C] [D]

Think carefully before you shade the answer spaces; erase completely any answers you wish to change.
Do all rough work on this question paper

Now answer the following questions

1. Correct 0.002468 to 3 significant figures.
   A. 0.002
   B. 0.003
   C. 0.00246
   D. 0.00247

2. Which of the following would possibly be the correct measurement of length of 26.05m?
   A. 26.048m
   B. 26.055m
   C. 26.044m
   D 25.052m

3. Which of the following is the largest divisor of 36 and 72?
   A. 2
   B. 6
   C. 12
   D. 36

4. Which of the following data set is Numeric and Discrete?
   A. Age
   B. Gender
   C. Height
   D. Number of persons living with HIV / AIDS

5. Which of the following is not a measure of central tendency?
   A. Mean
   B. Mode
   C. Median
   D. Range

6. A trader paid GH¢ 270.00 for a blender after a discount of 10% was allowed. Calculate the discount on the blender.
   A. GH¢ 27.00
   B. GH¢ 2.70
   C. GH¢ 30.00
   D. GH¢ 300.00

7. Ama is thrice as old as Yaw. How old is Yaw if Ama is 27 years old?
   A. 3 years
   B. 9 years
   C. 81 years
   D. 60 years

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8. The area of triangle is 30 cm^2. If the height is 6 cm, find the base of the triangle.
   A. 5 cm
   B. 10 cm
   C. 24 cm
   D. 180 cm

9. Find the diameter of a circle whose circumference is 44 cm?
   [Take $\pi = {}^{22}/_{7}$]
   A. 7 cm B. 14 cm
   C. 21 cm D. 28 cm

10. Which of the following is possibly the corresponding height of 3 persons, if the constart of proportionality is 3?
   A. 6 cm B. 12 cm
   C. 9 cm D. 27 cm

11. The gradient of a hill is 2 which of the following best describes the hill?
   A. Horizontal
   B. Vertical
   C. Slanted downwards toward the left
   D. Slanted downwards toward the right

12. The ratio of men to women in a stadium is 3 : 2, how many men were in the stadium on a day we have 1500 women?
   A. 750
   B. 4500
   C. 2250
   D. 3750

13. A cliff of height 6m has a bird on top of it. The distance between a boy on the ground and the bird is 10m. Find the distance between the foot of the cliff and the boy?
   A. 8m B. 4m
   C. 2m D. 16m

The relationship between the number of boys (x) and girls(y) in each class of a school is : $y = \frac{1}{2}x - 8$. Use the information above to answer questions 14 to 16.

14. If there were 20 boys in JHS 2, how many girls are in that class?
   A. 2 B. 3
   C. 5 D. 10

15. The girls in class 6 are 12, how many boys are in that class?
   A. 2 B. 20
   C. 30 D. 40

16. JHS 3 has 46 learners, how many are most likely to be boys
   A. 10 B. 26
   C. 36 D. 40

17. The mean of 3, 6, 6 and x is 7. Find the value of x?
   A. 7 B. 12
   C. 13 D. 20

18. The probability of obtaining a head in a biased toss of a coin is ${}^{2}/_{3}$, find the probability of obtaining a tail.
   A. $\frac{1}{2}$ B. $\frac{1}{3}$
   C. $\frac{2}{3}$ D. $\frac{4}{5}$

19. Amadu run around a tall circular shaped building. Which of the following best describes the locus of Amadu?
   A. The locus of Amadu equidistant from one fixed point
   B. The locus of Amadu equidistant from two fixed points
   C. The locus of Amadu equidistant from a straight line

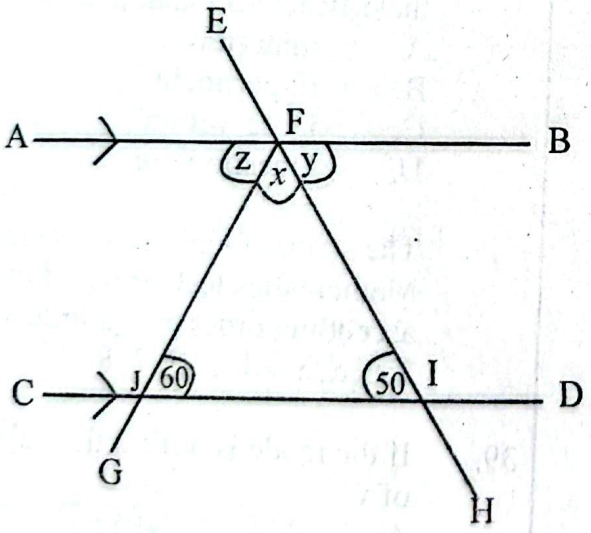
D. The locus of Amadu equidistant from a straight line

In JHS 3 class of Michel Camp 2 JHS , 28 learners study at least one of English or Mathematics. 32 study at most one of the two subjects. 20 study exactly one of the two subjects. Use the information above to answer question 20 - 22

20. How many learners study both English and Mathematics?
A. 8 B. 12
C. 48 D. 52

21. How many learners study neither English nor Mathematics?
A. 8 B. 12
C. 48 D. 52

22. How many learners are in JHS 3 of Micheal Camp 2 JHS?
A. 20 B. 28
C. 40 D. 52



In the diagram above AB is paralled to CD. Use the information above to answer questions 23 - 25

23. Find the value of x
A. 70°
B. 110°
C. 130°
D. 150°

24. Find the value of $x + y$
A. 70°
B. 110°
C. 120°
D. 140°

25. Find the value of z
A. 60°
B. 70°
C. 110°
D. 130°

26. For divisibility test for A the sum of the digits must be divisible by A. What is ?
A. 3 B. 2
C. 5 D. 9

27. The bearing of Amasaman from Nungua is 230°. Find the bearing of Nungua from Amasaman
A. 180°
B. 50°
C. 350°
D. 110°

28. If $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ and $\overrightarrow{CB} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$ find $\overrightarrow{AC}$

A. $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$ B. $\begin{pmatrix} 7 \\ 3 \end{pmatrix}$

C. $\begin{pmatrix} -1 \\ -11 \end{pmatrix}$ D. $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$

29. The perimeter of rectangle is 30 cm. If the length is 8 cm. Find the area of the rectangle.
A. 160 cm^2
B. 56 cm^2
C. 15 cm^2
D. 30 cm^2

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30. The point (-2, 6) is reflected in the line $y = x$, what is the image?
A. (-2, -6)
B. (-6, 2)
C. (6, -2)
D. (-6, -2)

31. How many lines of symmetry does a circle have?
A. 2
B. 1
C. 4
D. Infinite

32. Thelma and Eric shared a number of oranges in the ratio 4 : 3 respectively. Thelma got 30 more oranges than Eric. How many oranges did they share?
A. 12
B. 42
C. 210
D. 420

33. Factorize completely
$2ax - 6ay + bx - 3by$
A. $(2a + b)(x - 3y)$
B. $(x + 3y)(2a + b)$
C. $(x - 3y)(2a - b)$
D. $(x + 3y)(2a - b)$

34. Simplify $\frac{2a - b}{4a - 6b}$
A. $\frac{1}{2a - 3b}$
B. $\frac{1}{2}$
C. $\frac{2a - 3b}{2}$
D. $\frac{2}{2a - 3b}$

35. What is the ratio of the number of legs to head in a human body?
A. 1 : 2
B. 2 : 2
C. 2 : 1
D. 1 : 4

36. Mansa scored 12 out of 20 in a Mathematics test. What will be the angle representation of her score in a pie chart?
A. 36°
B. 72°
C. 216°
D. 240°

37. Ato Fynn spent $^1/_3$ of his time for the day on cooking. He further spent $^2/_5$ of the remaining time on sports. What fraction of the time did he send on sports?
A. $\frac{2}{5}$
B. $\frac{2}{15}$
C. $\frac{4}{15}$
D. $\frac{3}{5}$

38. Which of the following is the best way of collecting data on checking the efficacy of malaria medicine?
A. Interview
B. Experiment
C. Observation
D. Questionaire

The score of eight students in a Mathematics test arranged in ascending order are as followers:
$3, 3, x, x + 1, 5, 5, 7, 8$

39. If the mode is 5, find the value of x.
A. 3
B. 4
C. 5
D. 6

40. Find the mean score
A. 3
B. 4
C. 5
D. 6

END OF PAPER

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